

Isogeometric Analysis (Basis Functions)

Natural Coordinates, Derivatives, NURBS Derivatives

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Natural Coordinate System

The natural coordinate system is based on covariant basis vector:

$$\mathbf{g}_\alpha = \frac{\partial \mathbf{x}}{\partial \xi^\alpha}, \quad (1)$$

where ξ^α is the parametric space and that can be anything we prefer. We typically use $-1 \leq \xi^\alpha \leq 1$ in an element. Any parameter space is acceptable for dealing with element-based modeling. With the help of dual-basis system, we can have contravariant vectors such that

$$\mathbf{g}^\alpha \cdot \mathbf{g}_\beta = \delta^\alpha_\beta. \quad (2)$$

With that, the spatial derivative can be expressed as

$$\nabla = \frac{\partial}{\partial \mathbf{x}} = \mathbf{g}^\alpha \frac{\partial}{\partial \xi^\alpha}. \quad (3)$$

With that, the derivative of a function

$$f^h = \sum_{a \in \eta} f_a N_a \quad (4)$$

can be

$$\nabla f^h = \mathbf{g}^\alpha \sum_{a \in \eta} f_a N_{a,\alpha}. \quad (5)$$

Function Smoothness

C^d continuity refers to the smoothness of a function based on a common parameter space. Thus, essentially, our common space should be the knot space Ξ . Contrary, the expression G^d continuity can be used for local smoothness based on other parametrization. For example, we can see the smoothness between a pair of adjacent elements and that parametrization can be changed between other pair of adjacent elements. This is useful for non-trivial parametric spaces.

Implementation Directions

We introduce $\hat{\mathbf{e}}_\alpha$ for the basis vector in the parametric space, where this basis vectors are assumed to be orthonormal. The transformation from the parametric space to the physical space can be written as

$$\mathbf{Q} = \mathbf{g}_\alpha \hat{\mathbf{e}}_\alpha, \quad (6)$$

which is the counterpart of deformation gradient tensor. With that, the volume in parametric space $\hat{\Omega}$ to the physical space Ω can be converted as:

$$d\Omega = \underbrace{\det \mathbf{Q}}_{=J} d\hat{\Omega}. \quad (7)$$

Similar to Cauchy–Green tensor,

$$\mathbf{Q}^{-\top} \cdot \mathbf{Q} = g_{\alpha\beta} \hat{\mathbf{e}}_\alpha \hat{\mathbf{e}}_\beta. \quad (8)$$

Thus, J can be obtained by also:

$$J = \det^{\frac{1}{2}} [g_{\bullet\bullet}]. \quad (9)$$

For notational convenience, we introduce:

$$\underline{\mathbf{g}} = g_{\bullet\bullet}. \quad (10)$$

Implementation Directions

For integration:

1. Calculate $\mathbf{g}_\alpha = \sum_{a \in \eta} \mathbf{x}_a N_{a,\alpha}$
2. Calculate $g_{\alpha\beta} = \mathbf{g}_\alpha \cdot \mathbf{g}_\beta = [\underline{\mathbf{g}}]_{\alpha\beta}$
3. Calculate $J = \det^{\frac{1}{2}} \underline{\mathbf{g}}$

For the first derivatives:

1. Calculate $g^{\alpha\beta} = [\bar{\mathbf{g}}]^{\alpha\beta} = [\underline{\mathbf{g}}^{-1}]^{\alpha\beta}$
2. Calculate $\mathbf{g}^\alpha = g^{\alpha\beta} \mathbf{g}_\beta$

We note that $\mathbf{g}_\alpha$ is typically stored in the Cartesian basis system $\mathbf{e}_i$:

$$\mathbf{g}_\alpha \cdot \mathbf{e}_i = \underline{\mathbf{g}}_{\alpha i}, \quad (11)$$

$$\mathbf{g}^\alpha \cdot \mathbf{e}_i = \bar{\mathbf{g}}_{\alpha i}. \quad (12)$$

First Derivatives

We define the gradient of vector $\mathbf{u}$:

$$\nabla \mathbf{u} = \mathbf{g}^\alpha \mathbf{u}_{,\alpha}. \quad (13)$$

Because $\mathbf{u} = u_i \mathbf{e}_i$, this is the same as the scalar case:

$$\nabla \mathbf{u} = \mathbf{g}^\alpha \sum_{a \in \eta} u_{ai} N_{a,\alpha} \mathbf{e}_i, \quad (14)$$

where $u_{ai} = \mathbf{u}_a \cdot \mathbf{e}_i$.

The divergence operator is

$$\nabla \cdot \mathbf{u} = \nabla \mathbf{u} : \mathbf{I} \quad (15)$$

$$= \mathbf{g}^\alpha \cdot \mathbf{e}_i \sum_{a \in \eta} u_{ai} N_{a,\alpha} \quad (16)$$

$$= \bar{\mathbf{g}}_{\alpha i} \sum_{a \in \eta} u_{ai} N_{a,\alpha}. \quad (17)$$

Second Derivatives

The second derivative tensor is

$$\nabla\nabla u = \mathbf{g}^\alpha \left(\mathbf{g}^\beta u_{,\beta} \right)_{,\alpha} \quad (18)$$

$$= \mathbf{g}^\alpha \mathbf{g}^\beta u_{,\beta\alpha} + \mathbf{g}^\alpha \left(\mathbf{g}^\beta \right)_{,\alpha} u_{,\beta} \quad (19)$$

$$= \mathbf{g}^\alpha \mathbf{g}^\beta u_{,\beta\alpha} - \mathbf{g}^\alpha \mathbf{g}^\gamma \Gamma_{\alpha\gamma}^\beta u_{,\beta} \quad (20)$$

$$= \left(u_{,\alpha\beta} - \Gamma_{\alpha\beta}^\gamma u_{,\gamma} \right) \mathbf{g}^\alpha \mathbf{g}^\beta, \quad (21)$$

where

$$\Gamma_{\alpha\beta}^\gamma = \mathbf{g}^\gamma \cdot \mathbf{x}_{,\alpha\beta} = \mathbf{g}^\gamma \cdot \sum_{a \in \eta} \mathbf{x}_a N_{a,\alpha\beta} = \bar{\mathbf{g}}_{\gamma i} \sum_{a \in \eta} x_{ai} N_{a,\alpha\beta} \quad (22)$$

The Laplacian is

$$\Delta u = \mathbf{I} : \nabla\nabla u = \mathbf{g}^\delta \mathbf{g}_\delta : \nabla\nabla u = \left(u_{,\alpha\beta} - \Gamma_{\alpha\beta}^\gamma u_{,\gamma} \right) g^{\alpha\beta}. \quad (23)$$

If we only need Laplacian, we can prepare $\Gamma_{\alpha\beta}^\gamma g^{\alpha\beta}$.

NURBS

The NURBS basis functions are define by using B-spline functions N_a as follows:

$$R_a(t) = \frac{w_a N_a(t)}{\sum_{b \in \eta} w_b N_b(t)}, \quad (24)$$

where w_a is the NURBS weights. As can be seen that $w_a \leftarrow s w_a$ with $\forall s \neq 0$ resulting the same functions. If N_a has partition of unity, and $w_a = w_b$ with $\forall b \in \eta$, $R_a = N_a$. Obviously, the NURBS basis functions satisfy the partition of unity.

For notation convenience, we introduce the following functions:

$$Q_a(t) = w_a N_a(t), \quad (25)$$

$$S(t) = \sum_{b \in \eta} Q_b(t). \quad (26)$$

With that,

$$R_a(t) = \frac{Q_a(t)}{S(t)}. \quad (27)$$

First Derivative of NURBS Basis Functions

The first derivative of a NURBS basis functions is

$$R_a^{(1)} = \frac{Q_a^{(1)}S - R_aS^{(1)}}{S^2}. \quad (28)$$

Using $Q_a = SR_a$, we get

$$R_a^{(1)} = \frac{Q_a^{(1)} - Q_aS^{(1)}}{S}. \quad (29)$$

We note that we can derive this even simpler starting from:

$$Q_a^{(1)} = S^{(1)}R_a + SR_a^{(1)}. \quad (30)$$

Higher Order Derivative of NURBS Basis Functions

Using Leibniz rule, we can obtain d th derivative of Q_a as follows:

$$Q_a^{(d)} = \sum_{i=0}^d \binom{d}{i} S^{(i)} R_a^{(d-i)} \quad (31)$$

$$= S R_a^{(d)} + \sum_{i=1}^d \binom{d}{i} S^{(i)} R_a^{(d-i)}. \quad (32)$$

Thus,

$$R_a^{(d)} = \frac{Q_a^{(d)} - T_a^{(d)}}{S}, \quad (33)$$

and

$$T_a^{(d)} = \sum_{i=1}^d \binom{d}{i} S^{(i)} R_a^{(d-i)} \quad (34)$$

Derivatives of Higher Dimensional B-Splines Basis Functions

When a basis function is represented by a product form:

$$N_{\mathbf{a}}(\mathbf{t}) = \prod_{\alpha=1}^{n_{\text{pd}}} N_{a_{\alpha}}^{\alpha}(t^{\alpha}), \quad (35)$$

where $\mathbf{a} = \{a_1, \dots, a_{n_{\text{pd}}}\}$.

The d th derivative, which is $|\mathbf{d}| = d$, is denoted as:

$$N_{\mathbf{a}}^{(\mathbf{d})}(\mathbf{t}) = \prod_{\alpha=1}^{n_{\text{pd}}} (N_{a_{\alpha}}^{\alpha})^{(d_{\alpha})}(t^{\alpha}), \quad (36)$$

where $\mathbf{d} = \{d_1, \dots, d_{n_{\text{pd}}}\}$.

Serializing of Derivatives Combinations

The set of $|\mathbf{d}| = d$ is denoted as $\mathcal{D}_d$, and the size of the set is

$$\binom{d + n_{\text{pd}} - 1}{d}. \quad (37)$$

The idea of this comes from combination with repetition. Choose d times from the n_{pd} directions. We label $|\mathbf{d}| = d$ by binary number, such as “00010010”. “1” indicates that the separation of directions, which has $n_{\text{pd}} - 1$, and the number of “0” indicates the number of derivatives in that direction:

$$s = 18 = (0010010)_2 = 0\text{b} \underbrace{000}_{d_1=3} 1 \overbrace{00}^{d_2=2} 1 \underbrace{0}_{d_3=1}. \quad (38)$$

Upper parts corresponds to the lower number direction. With this system, we can count the numbers from $2^{n_{\text{pd}}} - 1$ to $2^d(2^{n_{\text{pd}}} - 1)$.

Serializing of Derivatives Combinations — Practice

- ▶ Make a table of $\mathcal{D}_d$ and label s for $n_{\text{pd}} = 1$ and $d = 2$
- ▶ Make a table of $\mathcal{D}_d$ and label s for $n_{\text{pd}} = 2$ and $d = 1$
- ▶ Make a table of $\mathcal{D}_d$ and label s for $n_{\text{pd}} = 2$ and $d = 2$
- ▶ Make a table of $\mathcal{D}_d$ and label s for $n_{\text{pd}} = 3$ and $d = 2$ and write a program to generate the table

Derivatives of Higher Dimensional NURBS Basis Functions

For $\mathbf{t} = (t^1, \dots, t^{n_{\text{pd}}})$,

$$Q_a(\mathbf{t}) = w_a N_a(\mathbf{t}), \quad S(\mathbf{t}) = \sum_{b \in \eta} Q_b(\mathbf{t}), \quad R_a(\mathbf{t}) = \frac{Q_a(\mathbf{t})}{S_a(\mathbf{t})}, \quad (39)$$

we express

$$R_a^{(\mathbf{d})} = \frac{R_a^{(\mathbf{d})} - T_a^{(\mathbf{d})}}{S}, \quad (40)$$

where

$$T_a^{(\mathbf{d})} = \sum_{\mathbf{i} \in \mathcal{T}_{\mathbf{d}}} \prod_{\alpha=1}^{n_{\text{pd}}} \binom{d_{\alpha}}{i_{\alpha}} R_a^{(\mathbf{d}-\mathbf{i})} S_a^{(\mathbf{i})}, \quad (41)$$

and

$$\mathcal{T}_{\mathbf{d}} = \{\mathbf{i} \in \mathbb{Z}^{n_{\text{pd}}} \mid i_{\alpha} = \{0, \dots, d_{\alpha}\} \setminus \{i_1 = \dots = i_{n_{\text{pd}}} = 0\}\}. \quad (42)$$

Here, we denoted

$$\mathbf{d} - \mathbf{i} = \{d_1 - i_1, \dots, d_{n_{\text{pd}}} - i_{n_{\text{pd}}}\}. \quad (43)$$

Implementation Directions

To calculate the higher order derivatives, we need lower order derivatives. Therefore, we store derivatives from $k = 0$, which is value, to d , which is the highest derivative required. The value is $\{N_a\}$ and the number is $n_{\text{en}} = |\eta|$. For d , we denote a given derivative $\mathbf{k}$ by

$$\mathcal{N}^{(\mathbf{k})} = \left\{ N_a^{(\mathbf{k})} \middle| a \in \eta \right\}, \quad (44)$$

which is size of n_{en} , and the total derivatives are

$$\mathcal{N}^k = \left\{ \mathcal{N}^{(\mathbf{k})} \middle| \mathbf{k} \in \mathcal{D}_k \right\}, \quad (45)$$

which is size of $\binom{d + n_{\text{pd}} - 1}{d}$. To calculate up to d th derivative, in total, we store

$$\mathcal{N} = \left\{ \mathcal{N}^k \middle| k = 0, \dots, d \right\}. \quad (46)$$

For efficiency, we typically store all the components in 1D, which is the total of

$$n_{\text{en}} \sum_{k=0}^d \binom{k + n_{\text{pd}} - 1}{k}. \quad (47)$$

Derivatives of Higher Dimensional NURBS Basis Functions — Practice

- ▶ Write down the formula for $n_{\text{pd}} = 2$ and $d = 2$
- ▶ Write a program to calculate derivatives for $n_{\text{pd}} = 2$ and $d = 2$